

Jarvis Auto Group

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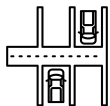
Executive Overview

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Problem Definition



→ **Misaligned Inventory & Demand**



→ **Excess Unsold Stock & Depreciation**



Key Decision Insights

- **What To Buy:** prioritizing which vehicles have the highest demand
- **What To Avoid:** which cars don't sell as much
- **When To Act:** anticipate demand shifts before they happen

Proposed Solution



→ **Inventory Analytics Framework**



→ **Agentic Recommendation Platform**

Dataset Overview & Requirements

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Dataset Overview & Transformations

Vehicle Market Sales Data (~600k Records, 80mb)



- **Vehicle, pricing, and sale details**
- Derived features, including price premium
- Removed outliers

Architecture Flow:

Raw Data → Data Cleaning & Transformations → EDA & Statistical Analysis → Business Insights

Data Quality & Production Considerations

Missing Values

Column	Missing (#)	Missing (%)
Transmission	65,352	11.69%
Body	13,192	2.36%
Other	44,829	8.02%

Backfill

Backfill
transmission and body
based on make/model

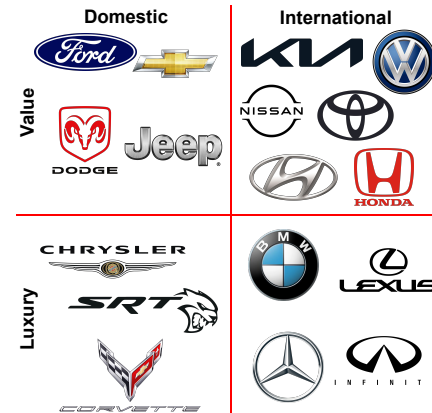
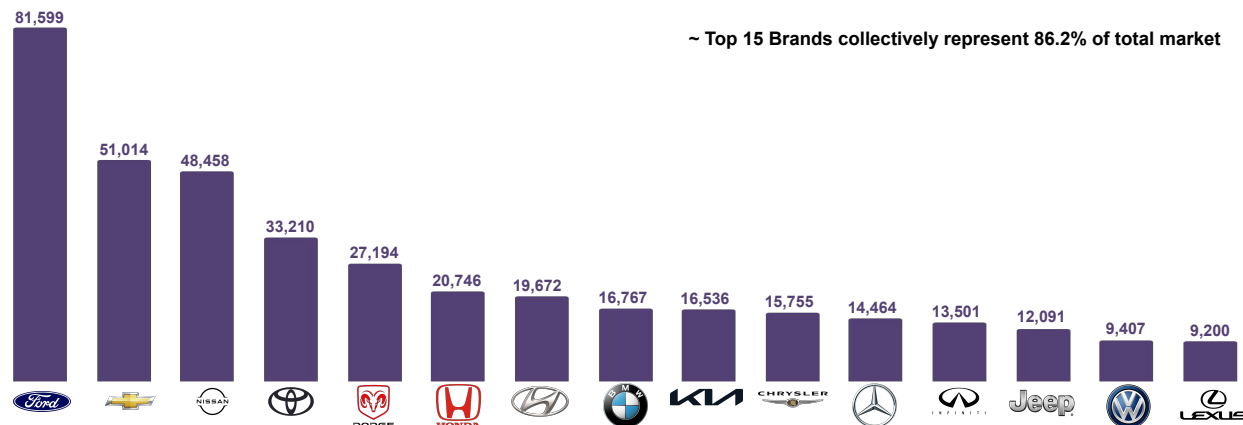
Before: 123,376 (~22%)
After: 68,856 (~12%)

- **Scalable, dashboard-ready analytics workflow.**

Analytical Insights & Findings I

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Top Vehicle Brands by Sales Volume



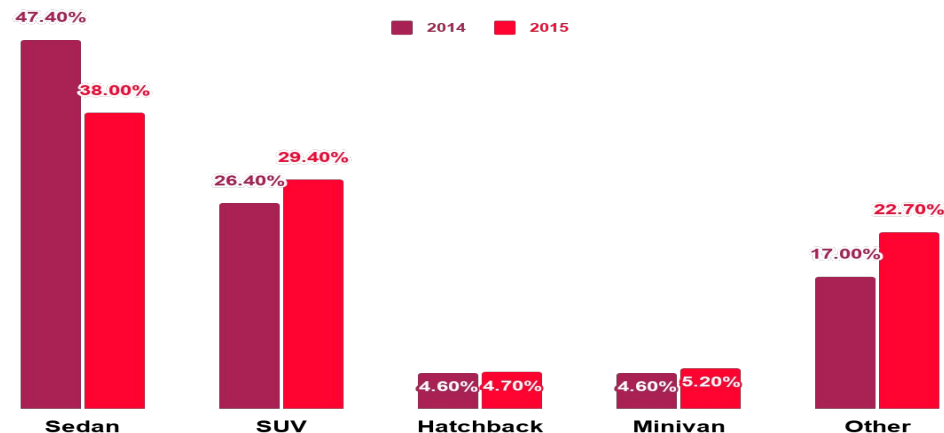
Insights

- The vehicle market is **highly concentrated**, **top 15 brands = 86.2% of total sales volume**
- **brand groups: value & luxury domestic manufacturers, value-focused international brands, and premium luxury automakers**
- **Top 15 brands unchanged in 2025**

Analytical Insights & Findings II

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Shifting Consumer Preferences Across Vehicle Body Types



Sedan: Demand drops **-22%** YoY



SUV: **+10.7%** increase in sales volume



Hatchback: Mostly Unchanged **+2.2%**



Minivan: Significant growth **+12.2%**



Other*: Rapid growth **+28.7%**

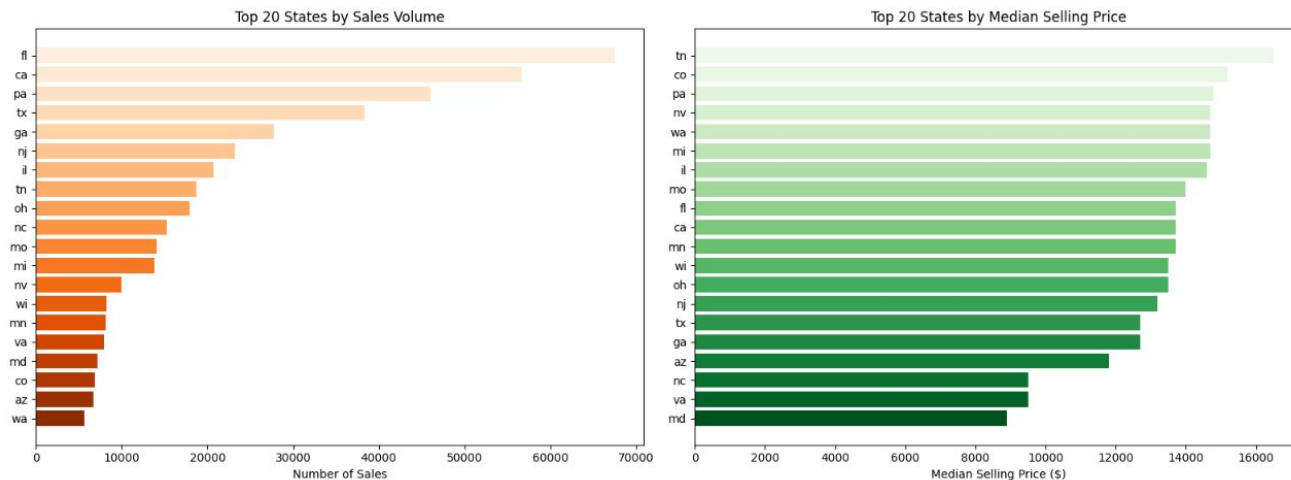
Insights

- SUVs, Minivans, and Other experience **double digit growth**; consumer shift away from traditional sedans and more towards utility focused body types
- Today: SUVs (52 – 55%), Sedans (15 – 25%)*

Analytical Insights & Findings III

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Geographic Insights



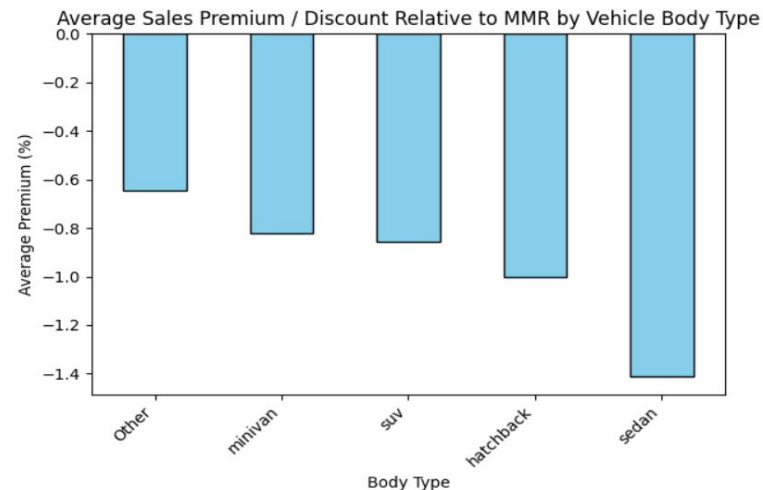
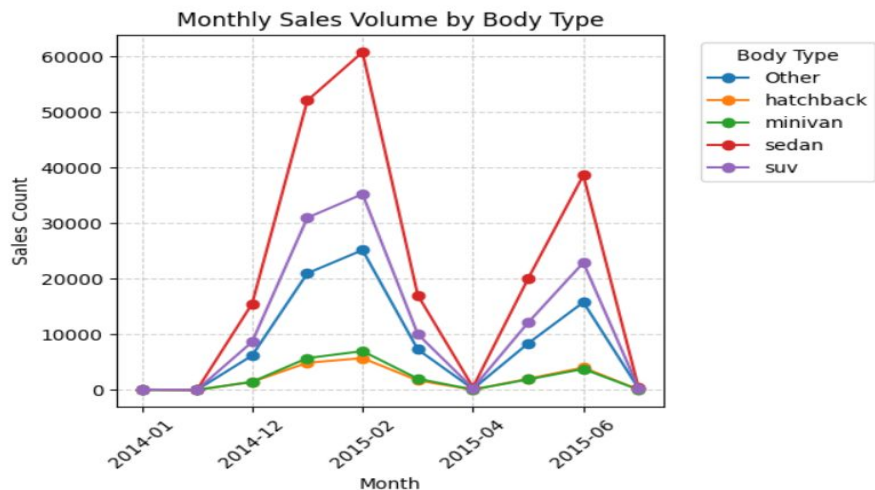
Insights

- **Florida:** Largest auction market by volume; California and Pennsylvania next.
- **Tennessee & Colorado:** Higher luxury vehicle concentration; higher median selling prices.
- **2025*:** similar volumes in dominant states; higher median prices in AK, AR & WY.

Analytical Insights & Findings IV

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Average Monthly Sales Volume & Average Sales Premium/ Discount relative to MMR

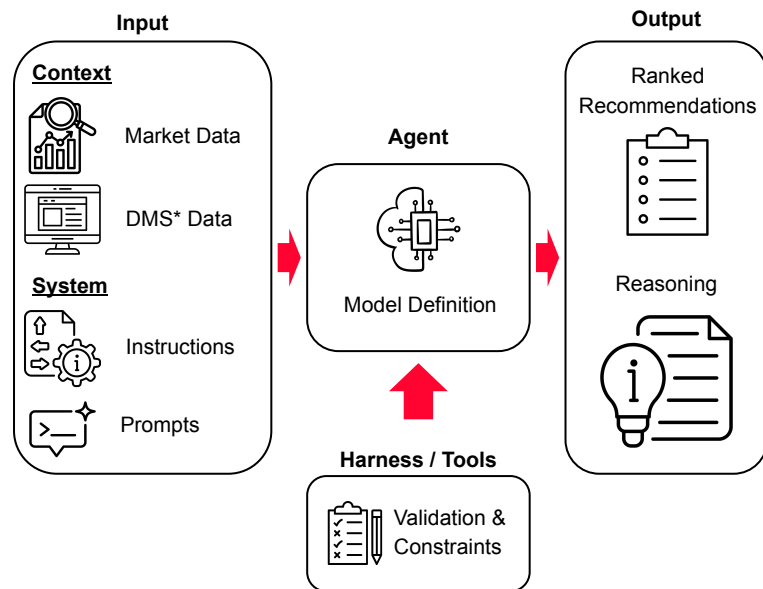


- **Sedans:** Highest sales volume across 2014–2015; sales decline across all segments from February to April.
- **Sedans:** Largest discount to MMR (~1.4% below); **Minivans & SUVs:** Closest to MMR (~0.8% below).
- **2026:** Jan–Feb lowest sales; early March dip; Sedans (Highest Depreciation); SUVs (Lowest Value Retention); Minivans (Strongest Value Retention)

Agentic Inventory Intelligence Platform

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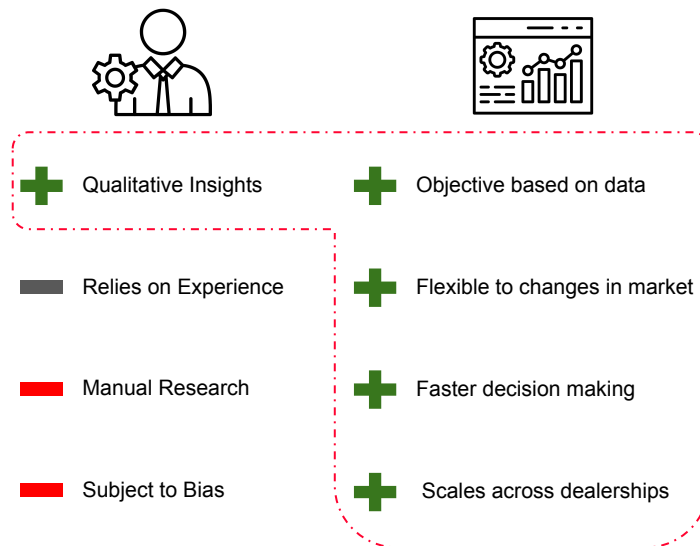
Proposed Solution Overview



➔ **Output purchase recommendations with data backed reasoning and validation**

Objective Outcome

Create a data-driven approach for inventory purchasing decisions that maximizes revenue and profitability



Key Takeaways & Next Steps

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Key Takeaways

Recommended Action

Future Roadmap



Market concentration is high



Concentrate on Core 15 Brands



Demand shift to utility vehicles



Shift Inventory Towards Utility Vehicles



Regional market dynamics



Focus Sourcing on Volume Markets
(FL & CA)



Sedans underperform on
value retention



Target Vehicles Closest to MMR

Data Infrastructure Automation

Real Time Pricing Optimization

Agentic Inventory
Recommendation Platform

Questions
